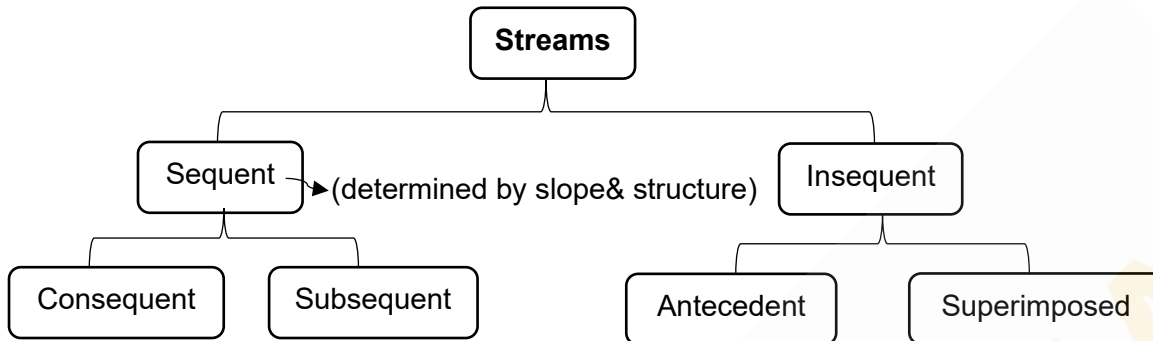


INDIAN DRAINAGE SYSTEM

→ **Drainage** -- flow of water through a fixed channel.

→ **Drainage system** – gives an account of origin and evolution of rivers & their tributaries.

→ **Drainage Pattern** – tells about arrangement/shape which river & tributaries make.



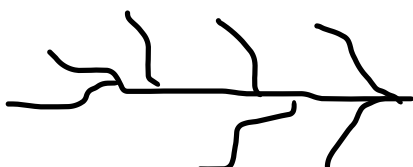
→ Parameters that determine the stream:

- Slope
- Geological structure
- Source
- Time of origin
- Nature of landform

- (i) **Consequent stream :** stream follow slope of land over which it is formed.
Also called as Dip streams.
Ex. Godavari & krishna
- (ii) **Subsequent stream:** streams originated after consequent stream and follows the geological structure.
Ex. Chambal, ken, betwa
- (iii) **Antecedent stream:** originated before the upliftment of the land surface.
Maintains its original course & pattern.
Ex. Himalayan rivers
- (iv) **Superimposed stream:** A river inherited its previous form of geological structure though its flowing on a newly formed geological structure.
Ex. Damodar, Banad, Subarnarekha

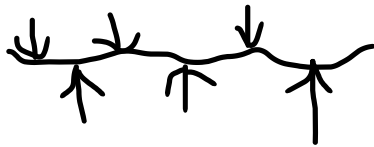
Drainage patterns:

- (i) **Dendritic / Pinnate pattern:**



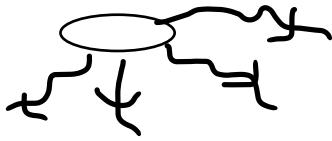
Appears like irregular tree branch
Develops in region with uniform lithology & insignificant faulting process.
Ex. Indus, Godavari, ganga

(ii) Trellis Pattern :



short subsequent stream meet master stream at right angle.
Ex. Chotanagpur plateau region

(iii) Radial pattern:



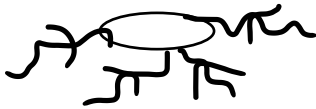
Tributaries move along the slopes of the structure and drain in all direction.

Ex. Streams of Saurashtra region, narmada, son, jhokila

(iv) Parallel pattern:

tributaries move in parallel direction in a uniform slopy region.
Ex. Rivers draining into Arabian sea.

(v) Centripetal pattern:



streams converge from all directions in low laying basin.
Ex. Streams of Ladakh, Tibet.

Obsequent stream: streams where tributaries join the aster stream in opposite direction.

Peninsular river	Himalayan river
Perennial in nature	Seasonal in nature
Long length of river	Shorter length of river
River basin size is large	River basin size is shorter
River regimes – two maxima	River regimes – one maxima
High erosion → deeper valleys	Lesser erosion → shallow valleys
High meandering paths	Straight courses
Forms large deltas	

LANDFORMS AND AGENTS

River	Wind	Sea-waves	Glaciers
Potholes	Ripple marks	Cliffs	Cirque / corrie
waterfalls	Sand dunes	Sea caves	Hanging valleys
Meanders	Barchans	Sea arches & stacks	Horns & aretes
Ox-bow lakes	Loess	Wave cut platforms	Moraines
V shaped valley	Piedplains	Beaches	Eskers
Alluvial fans	Deflation hollows	Spits & Tombolos	Drumlins
Flood plain	Zeugen	Barrier islands	
Levees	Yardangs	Deltas	
Braided channels	Demoiselles	Estuaries	
Deltas	Inselbergs	Lagoons	
	Mushroom tables	fjords	